

Endogenous Reachability Collapse and Recovery-Constrained Dynamics (ERC-R): A Minimal Interpretation in Oxygen-Related Redox Systems

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AI-assisted modeling and analysis using ChatGPT and Grok
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Abstract

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We propose a minimal interpretation of recoverability constraints in dynamical systems in the context of oxygen-related redox processes. Building on prior minimal models in which recoverability depends on the temporal structure of perturbations, we suggest that similar constraints may arise in systems where recovery is governed by finite internal timescales.

In biological systems, oxygen-mediated redox dynamics involve tightly regulated kinetics that coordinate electron transfer, buffering, and dissipation. Under this perspective, system failure may emerge when imposed perturbations exceed the capacity of these processes to track temporal structure within their intrinsic timescales, even when total load remains within nominal limits but its temporal structure exceeds internal recovery capacity.

This interpretation does not assume a specific mechanism, but highlights the potential role of redox dynamics as constraints on the temporal accessibility of functional states.

1. Introduction

In minimal dynamical systems, recoverability is often interpreted as the ability of a system to return to a baseline state following perturbation.

Previous work (ERC-D and ERC-S) has shown that recoverability may depend not only on the magnitude of perturbations, but on their temporal structure. In these models, sharp or probabilistic boundaries emerge in parameter space, separating regimes of recovery from regimes of failure.

The present work does not introduce a new dynamical model, but instead proposes an interpretation: that similar constraints may arise in biological systems where recovery is governed by finite internal timescales.

The goal is not to establish a specific biochemical mechanism, but to explore whether oxygen-related redox processes may operate as dynamical constraints on recoverability.

This perspective is consistent with the possibility that functional states remain present but become dynamically inaccessible when recovery processes fail to resolve perturbations within finite time windows.

2. Conceptual Framework

The observed dependence of recovery on temporal structure, rather than solely on total input magnitude, may be consistent with systems in which recovery is constrained by finite internal timescales.

In biological contexts, oxygen-mediated redox processes are known to involve tightly regulated kinetics; however, in this work they are considered as a minimal interpretative framework rather than a demonstrated mechanism, where restoration of redox balance requires coordinated:

- electron transfer
- buffering capacity
- dissipative processes

Under this perspective, the system may be interpreted as one in which:

- perturbations impose a transient demand on internal recovery processes
- recovery depends on whether these processes can track the imposed temporal structure within their intrinsic timescales
- failure may emerge when recovery processes cannot resolve perturbations within their intrinsic timescales, even if total load remains within nominal limits

This interpretation does not assume a specific mechanism, but suggests that oxygen-related redox dynamics may act as constraints on the temporal accessibility of functional states, rather than solely as drivers of metabolic activity.

3. Minimal Interpretation of Recoverability

Within this framework, recoverability may be understood as a dynamical condition constrained by finite temporal resolution, rather than a static property of the system.

A system may remain structurally capable of recovery, while losing the ability to access that state dynamically within a finite observation window.

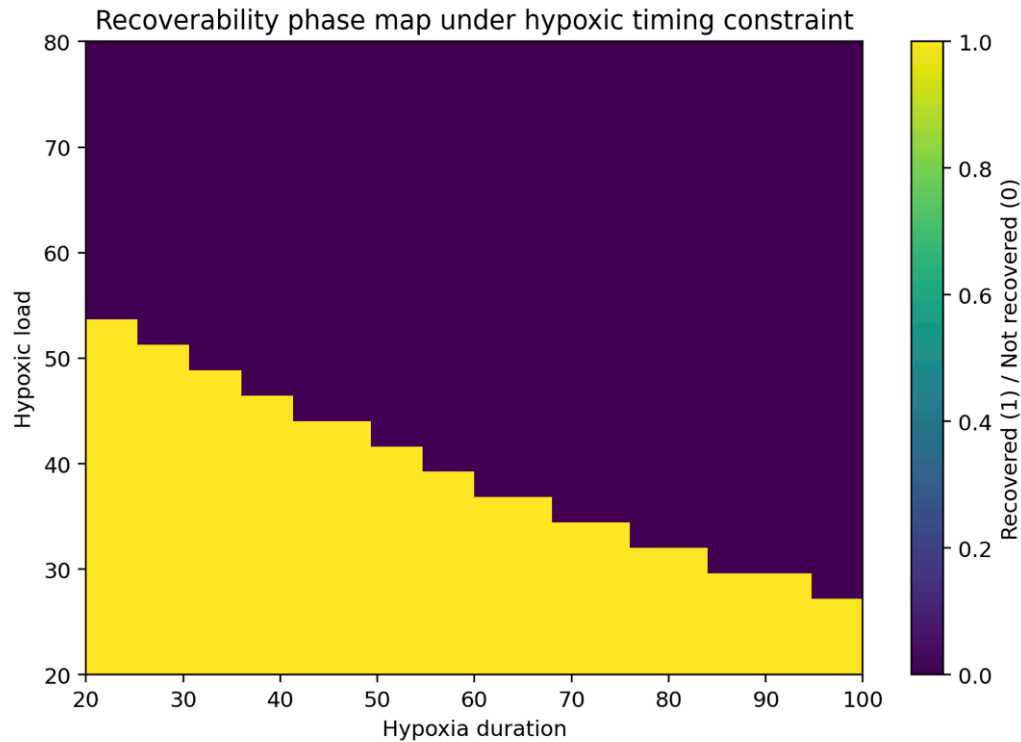
This perspective is consistent with the possibility that:

- functional states remain present in the state space
- but become dynamically inaccessible under certain temporal conditions

Such behavior aligns with prior minimal results, where recoverability boundaries emerge without removal of states.

4. Illustration: Hypoxic Timing Constraint

Figure 1. Recoverability phase map under hypoxic timing constraint in a minimal dynamical system.



The figure shows the classification of simulated trajectories as recovered or not recovered as a function of hypoxic load and hypoxia duration.

A clear boundary emerges between regimes in which the functional capacity variable returns toward baseline and regimes in which recovery fails within the observation window.

The boundary is inclined rather than axis-aligned, suggesting that recoverability depends on the joint configuration of load and duration rather than on either variable alone.

This behavior is consistent with the possibility that prolonged perturbations may impair return by sustaining internal constraint, even when peak load is lower than in shorter exposures.

This pattern is consistent with a regime in which recoverability is limited by the ability of internal processes to resolve temporal demand, rather than by peak load alone.

Dynamic proximity to collapse

In addition to the phase map representation, the results may be interpreted in terms of a dynamic proximity signal.

In the minimal deterministic model (ERC-D), a time-dependent safety margin $S(t)$ was defined to capture the distance between system trajectories and the recoverability boundary.

Under this perspective, the hypoxic regime boundary shown here may not only represent a static separation in (load, duration) space, but could also be associated with a progressive reduction in dynamic accessibility during the perturbation.

In biological terms, this may correspond to a situation in which oxygen-related redox processes gradually lose the ability to resolve temporal demand, even before the system fully crosses into a non-recoverable regime.

This suggests that loss of recoverability may be preceded by a continuous degradation of dynamic margin, rather than occurring as an abrupt transition.

5. Relation to Prior Work

The figure shows the classification of simulated trajectories as recovered or not recovered as a function of hypoxic load and hypoxia duration.

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6. Falsifiable Implications

The proposed interpretation leads to the following testable implications:

1. Dependence on temporal structure
If recovery depends only on total perturbation magnitude and not on temporal structure, this framework would be challenged.
2. Mismatch between load and outcome
Systems exposed to equal total load but different temporal structures may exhibit different recovery outcomes.
3. Finite-timescale limitation

Failure may occur when recovery processes cannot resolve perturbations within their intrinsic timescales.

4. Persistence of states

5. If failure requires removal of functional states rather than loss of accessibility, the interpretation would not hold

7. Discussion

These results suggest that oxygen-related redox dynamics may act not only as mediators of energy transfer, but potentially as constraints on system dynamics, although this work does not establish a specific mechanistic role and remains at the level of a minimal dynamical interpretation.

In this view, the relevant variable may not be total oxidative load, but whether recovery processes can track temporal demand within finite time windows.

This interpretation is intentionally minimal and does not imply that oxygen is unique in this role, but that it provides a tractable case for exploring recovery-constrained dynamics.

The model remains intentionally minimal and does not assume a specific biological mechanism. Further work is required to determine whether similar constraints operate in real systems.

8. Conclusion

We propose that recoverability in dynamical systems may be constrained by the ability of internal processes to resolve perturbations within finite timescales.

In the context of oxygen-related redox dynamics, this suggests that system failure may arise not from the removal of functional states, but from loss of dynamic accessibility.

This interpretation provides a minimal conceptual bridge between abstract dynamical models and biological systems, and may serve as a basis for future experimental investigation.

This work extends the deterministic framework introduced in ERC-D, where a sharp recoverability boundary emerges in a minimal system.